Project Report

Automatic Plant Watering System with Manual Control

Student Name: Dabojit Talukder Shopno
Department: Electronics Engineering
Institution: Chittagong Polytechnic Institute

1. Introduction

This project aims to solve the problem of inconsistent plant watering in agriculture and home gardening. By integrating a soil moisture sensor with an automated pump, the system ensures plants receive the exact amount of water needed. Additionally, a manual override feature allows users to control the system via a computer, adding a layer of professional flexibility.

2. System Components

Component	Description
Arduino Uno	Main microcontroller (brain) for processing logic.
Soil Moisture Sensor	Measures the water content in the soil in real-time.
5V Relay Module	Acts as a bridge to safely switch the high-power water pump.
Submersible Water Pump	Transfers water from the reservoir to the soil.

Component	Description
3000mAh Li-ion Battery	Dedicated high-capacity power source for the pump.

3. Working Principle

A. Automatic Mode

The system constantly monitors the soil moisture level. When the moisture drops below a predefined threshold (e.g., 600), the Arduino activates the relay, which turns on the pump. Once the moisture level rises, the pump automatically deactivates.

B. Manual Override (IoT Capability)

The user can input commands via the Serial Monitor. Typing "ON" will trigger a 5-second watering burst regardless of the sensor reading. This is essential for testing or providing extra water when the user deems necessary.

4. Power Management Strategy

A "Dual Power" design is implemented to ensure stability. The Arduino is powered by the laptop's USB, while the pump uses a separate 3000mAh Lithium battery. This prevents the "system hang" issue caused by voltage spikes when the motor starts.

5. Implementation Code

```
void loop() {
  if (Serial.available() > 0) {
    String input = Serial.readStringUntil('\n');
    input.trim();
    if (input == "ON") {
      digitalWrite(relayPin, LOW);
      delay(5000); // 5-second burst
      digitalWrite(relayPin, HIGH);
    }
  }
  // Auto Mode Sensor Logic...
```

```
  int value = analogRead(sensorPin);
  if (value > threshold) { digitalWrite(relayPin, LOW); }
  else { digitalWrite(relayPin, HIGH); }
}
```